IS 607 Assignment - Tidying and Transforming Data

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```
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(knitr)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
# I have Used the direct raw GitHub URL
data_url <- "https://raw.githubusercontent.com/Muhammad-Imran91/607/main/data.csv"

raw_df <- read_csv(data_url, show_col_types = FALSE)
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
# Standardize or rename first column if needed (e.g., if first column is named `...1` or `X1`)
if (!"Airline" %in% colnames(raw_df)) {
  raw_df <- raw_df %>% rename(Airline = 1)

}
```

```r
kable(raw_df, caption = "Folloing is a Raw Untidy Input Data")
```

Table 1: Folloing is a Raw Untidy Input Data

Airline	Status	Los Angeles	Phoenix	San Diego	San Francisco	Seattle
ALASKA	on time	497	221	212	503	1841
NA	delayed	62	12	20	102	305
NA	NA	NA	NA	NA	NA	NA
AM	on time	694	4840	383	320	201
WEST						
NA	delayed	117	415	65	129	61

```r
cleaned_wide_df <- raw_df %>%
  # Filter out completely empty rows
  filter(!if_all(everything(), is.na)) %>%
  # Fill missing Airline names downward
  fill(Airline, .direction = "down") %>%
  # Parse numeric columns formatted with commas
  mutate(across(c(`Los Angeles`, Phoenix, `San Diego`, `San Francisco`, Seattle),
                ~ as.numeric(gsub(",", "", .))))

kable(cleaned_wide_df, caption = "Following is a Cleaned Wide Data Structure")
```

Table 2: Following is a Cleaned Wide Data Structure

Airline	Status	Los Angeles	Phoenix	San Diego	San Francisco	Seattle
ALASKA	on time	497	221	212	503	1841
ALASKA	delayed	62	12	20	102	305
AM	on time	694	4840	383	320	201
WEST						
AM	delayed	117	415	65	129	61
WEST						

```r
flights_long <- cleaned_wide_df %>%
  pivot_longer(
    cols = c(`Los Angeles`, Phoenix, `San Diego`, `San Francisco`, Seattle),
    names_to = "City",
    values_to = "Count"
  ) %>%
  pivot_wider(
    names_from = Status,
    values_from = Count
  ) %>%
  rename(
    on_time = `on time`,
    delayed = delayed
  ) %>%
  mutate(
    total_flights = on_time + delayed,
    delay_rate = delayed / total_flights,
    on_time_rate = on_time / total_flights
```

```
  )

kable(flights_long, caption = "Tidied Long Data Format with Delay Rates")
```

Table 3: Tidied Long Data Format with Delay Rates

Airline	City	on_time	delayed	total_flights	delay_rate	on_time_rate
ALASKA	Los Angeles	497	62	559	0.1109123	0.8890877
ALASKA	Phoenix	221	12	233	0.0515021	0.9484979
ALASKA	San Diego	212	20	232	0.0862069	0.9137931
ALASKA	San Francisco	503	102	605	0.1685950	0.8314050
ALASKA	Seattle	1841	305	2146	0.1421249	0.8578751
AM WEST	Los Angeles	694	117	811	0.1442663	0.8557337
AM WEST	Phoenix	4840	415	5255	0.0789724	0.9210276
AM WEST	San Diego	383	65	448	0.1450893	0.8549107
AM WEST	San Francisco	320	129	449	0.2873051	0.7126949
AM WEST	Seattle	201	61	262	0.2328244	0.7671756

```
overall_summary <- flights_long %>%
  group_by(Airline) %>%
  summarise(
    total_on_time = sum(on_time),
    total_delayed = sum(delayed),
    total_flights = sum(total_flights),
    overall_delay_rate = sum(delayed) / sum(total_flights),
    overall_on_time_rate = sum(on_time) / sum(total_flights)
  )

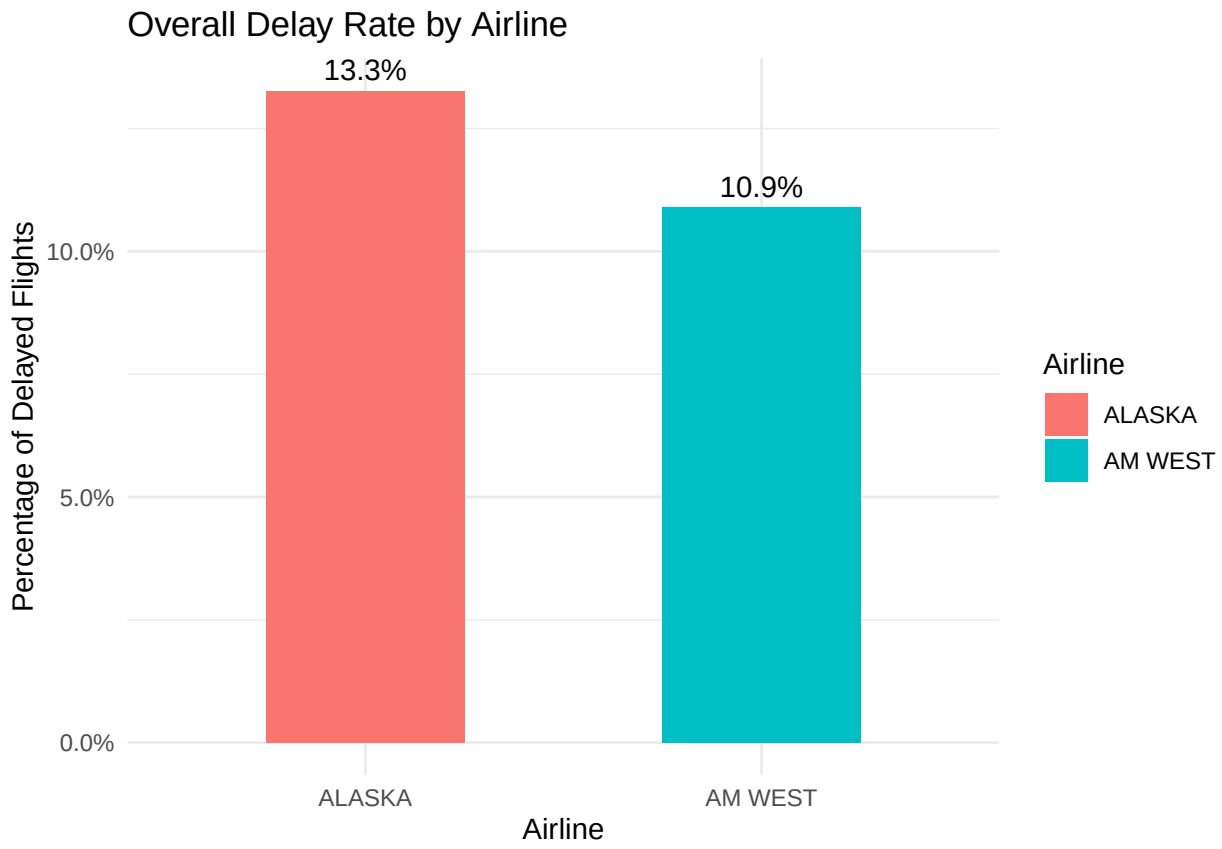
kable(overall_summary,
      digits = 4,
      col.names = c("Airline", "On-Time Count", "Delayed Count", "Total Flights", "Overall Delay Rate",
      caption = "Below is an Overall Flight Performance Summary")
```

Table 4: Below is an Overall Flight Performance Summary

Airline	On-Time Count	Delayed Count	Total Flights	Overall Delay Rate	Overall On-Time Rate
ALASKA	3274	501	3775	0.1327	0.8673
AM WEST	6438	787	7225	0.1089	0.8911

```
ggplot(overall_summary, aes(x = Airline, y = overall_delay_rate, fill = Airline)) +
  geom_col(width = 0.5) +
  scale_y_continuous(labels = percent_format(accuracy = 0.1)) +
  labs(title = "Overall Delay Rate by Airline",
       y = "Percentage of Delayed Flights",
```

```
       x = "Airline") +
  theme_minimal() +
  geom_text(aes(label = percent(overall_delay_rate, accuracy = 0.1)), vjust = -0.5)
```




```
city_summary <- flights_long %>%
  select(Airline, City, delayed, total_flights, delay_rate)

kable(city_summary,
      digits = 4,
      col.names = c("Airline", "City", "Delayed Count", "Total Flights", "Delay Rate"),
      caption = "Delay Rate Breakdown by City")
```

Table 5: Delay Rate Breakdown by City

Airline	City	Delayed Count	Total Flights	Delay Rate
ALASKA	Los Angeles	62	559	0.1109
ALASKA	Phoenix	12	233	0.0515
ALASKA	San Diego	20	232	0.0862
ALASKA	San Francisco	102	605	0.1686
ALASKA	Seattle	305	2146	0.1421
AM WEST	Los Angeles	117	811	0.1443
AM WEST	Phoenix	415	5255	0.0790
AM WEST	San Diego	65	448	0.1451
AM WEST	San Francisco	129	449	0.2873
AM WEST	Seattle	61	262	0.2328

```r
ggplot(flights_long, aes(x = City, y = delay_rate, fill = Airline)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = percent_format(accuracy = 0.1)) +
  labs(title = "Delay Rate by Airline Across 5 Cities",
       y = "Percentage of Delayed Flights",
       x = "Destination City") +
  theme_minimal() +
  geom_text(aes(label = percent(delay_rate, accuracy = 0.1)),
            position = position_dodge(width = 0.9), vjust = -0.5, size = 3)
```

